

Take-Home Assignment – Clinical BERT Real-Time Inference API

Role: Data Engineer

Objective: Build and deploy a real-time inference API for clinical text classification using Hugging Face and Google Cloud/AWS.

Background

Healthcare systems often contain unstructured clinical notes. Understanding the **assertion status** (e.g., PRESENT, ABSENT, CONDITIONAL) of medical concepts in text is critical for downstream analytics and diagnostics.

You will deploy the Hugging Face model [bvanaken/clinical-assertion-negation-bert](https://huggingface.co/bvanaken/clinical-assertion-negation-bert) as a real-time API that classifies sentences with a label and confidence score.

Core Requirements

1. Build the API

- Framework: **FastAPI** or **Flask**
- Use Dockerfile based on python:3.12-slim

Endpoint:

```
POST /predict
```

Input (JSON):

```
{  
  "sentence": "The patient denies chest pain."  
}
```

Output (JSON):

```
{  
  "label": "ABSENT",  
}
```

```
"score": 0.9842
}
```

2. Load and Cache the Model

Use Hugging Face Transformers to load the model:

`bvanaken/clinical-assertion-negation-bert`

- Load the model only once during startup (not per request).

3. Performance Expectation

- API response time: < **500ms** for short clinical sentences.

Cloud Deployment – GCP (Free Tier)

Deploy your service using **Google Cloud Run**, which supports Dockerized APIs.

- Use your free credits (\$300 for new users)
- Containerize your API using **Docker**
- Deploy to **Cloud Run** (or Cloud Functions if preferred)
- Use **Artifact Registry** for container storage

CI/CD with GitHub Actions (Bonus)

Suggested set up GitHub Actions/ Can plan your own:

1. `ci.yml` – Continuous Integration

- Trigger: PRs and pushes to `main`
- Steps:
 - Install dependencies
 - Run lint checks (e.g., `black`, `flake8`)
 - Run unit tests

2. `cd.yml` – Continuous Deployment

- Trigger: Push to `main`
- Steps:
 - Build and push Docker image to GCP Artifact Registry or AWS

- Deploy to Cloud Run using `gcloud` CLI

Test Cases

Please ensure your API can handle these:

Sentence	Expected Label
"The patient denies chest pain."	ABSENT
"He has a history of hypertension."	PRESENT
"If the patient experiences dizziness, reduce the dosage."	CONDITIONAL
"No signs of pneumonia were observed."	ABSENT

Suggested Project Structure

```
clinical-bert-api/
├── app/
│   ├── main.py           # FastAPI/Flask app
│   ├── model.py          # Model loading & prediction logic
│   └── schemas.py        # Input/output validation
├── tests/
│   └── test_api.py       # Basic test cases
├── Dockerfile
├── requirements.txt
├── .github/
│   └── workflows/
│       ├── ci.yml
│       └── cd.yml
├── README.md
└── deploy.sh             # Optional GCP deploy script
```

Submission Instructions

Please provide:

- GitHub repo with:
 - Source code
 - `README.md` with instructions
 - Dockerfile
 - CI/CD workflows
 - API URL (deployed on cloud like AWS/GCP)
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README Must Include

- Project overview
- Setup instructions for local development
- How to deploy on your preferred cloud.
- Example API usage using python
- Any known issues or tradeoffs

Extra Credit (Optional)

- Add a `/health` endpoint for a health check
- Support batch predictions (multiple sentences)
- Auto-deploy on tag creation.